Humanoid Underactuated Juggling via MPC Control and Energy-shaping

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INTRODUCTION & MOTIVATION

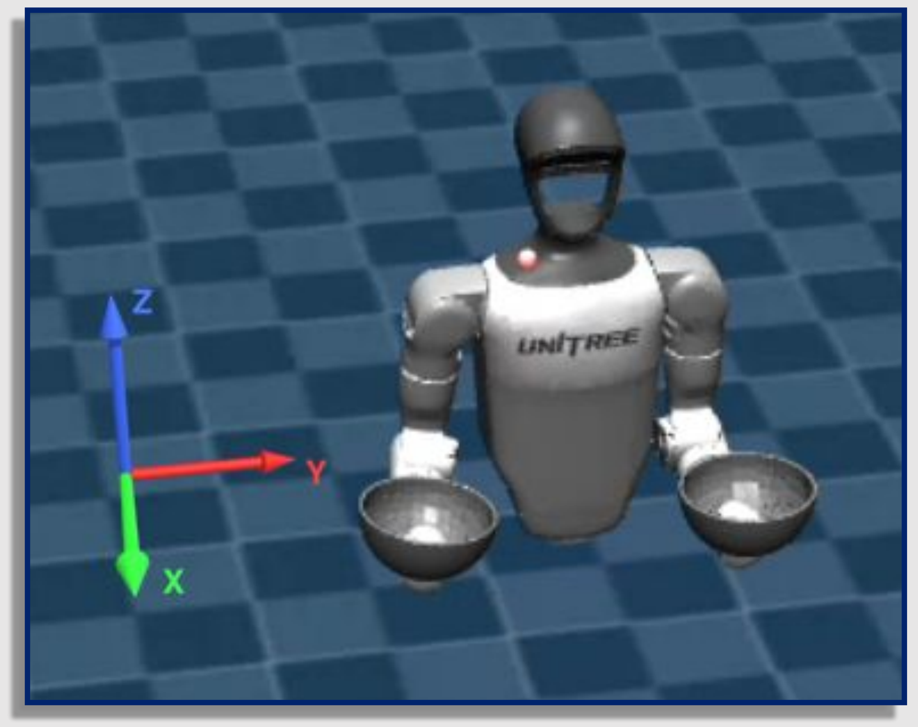
- An emerging niche for commercial robotics is humanoid interactions with ***household items***.
- Modern robotics still struggles with ***dynamic***, ***contact-rich*** tasks that humans can perfect.
- For robust safety precautions in human settings, these robots must be able to ***adapt*** mid-task, even under ***unexpected constraints***.
- We explore ***underactuated juggling*** as an example scenario where a humanoids' end-effectors are "busy" holding a household item.
- Constrained to ***bowls*** as end-effectors, it must ***juggle balls*** while ***avoiding obstacles***.
- Implemented on a ***Unitree G1*** via a ***MuJoCo*** simulation, we account for actuator limits and resistances throughout each arm's 7 DoF.

BACKGROUND

- The ***Energy-shaping*** controller segments the throwing action into ***3 phases***:
  1. ***Spin***: Bowl stays upright and builds ball speed
  2. ***Tilt***: Bowl angles toward the throw direction
  3. ***Release***: Bowl flips to let the ball exit naturally.
- ***Each phase*** has its own ***simplified cost***, with the controller using the negative of that cost as the phase reward.

Bowl position
Actuation energy input
Bowl orientation
Actuation near limits

Video Results



METHODS

- We utilise and compare various control methods, including ***MPC***, and ***Energy-shaping*** for discretized throwing stages.
- We break down motion into ***3 stages***:
  - ***Ball catching*** (MPC tracking X-Y trajectory)
  - ***End-effector return*** to initial position (MPC)
  - ***Ball throwing*** (MPC or Energy-shaping).
- Cost functions for throwing utilise a bowl's instantaneous cartesian (and angular) ***position and velocity***, as well as the target ball's such states upon the instant of catching.
- Costs also incur at rapid ***actuation*** and operation near ***limits***.
- MPC throwing costs:

$$\ell_{\text{throw}} = w_p \left\| p_a - p_a^{\text{des}}(t) \right\|^2 + w_v \left(v_a^{\text{des}}(t) - \bar{v}_a \right)^2 + w_{\text{ball}} \mathbb{I}_{\text{not release}} \left\| p_b - p_a \right\|^2 + w_l L + w_u U$$

- Energy-shaping throwing costs:
  1. $$\ell_{\text{spin}} = w_{\text{pos}}^{\text{spin}} \left\| \mathbf{p}_a - \mathbf{p}_a^0 \right\|^2 + w_z^{\text{spin}} \left(z_a - z_a^0 \right)^2 + w_{\text{upright}}^{\text{spin}} \left(1 - \hat{\mathbf{z}}_a^\top \mathbf{e}_z \right)^2 + w_\omega^{\text{spin}} \left(\dot{q}_{\text{wrist}} - s_a \omega_{\text{spin}} \right)^2 + w_l L + w_u U$$
  2. $$\ell_{\text{tilt}} = w_{\text{pos}}^{\text{tilt}} \left\| \mathbf{p}_a - \mathbf{p}_a^0 \right\|^2 + w_z^{\text{tilt}} \left(z_a - z_a^0 \right)^2 + w_{\text{tilt}} \left\| \hat{\mathbf{z}}_a - \hat{\mathbf{z}}_a^{\text{tilt}} \right\|^2 + w_\omega^{\text{tilt}} \left(\dot{q}_{\text{wrist}} - \alpha \omega_{\text{spin}} \right)^2 + w_l L + w_u U$$
  3. $$\ell_{\text{rel}} = w_{\text{rel}} \left\| \hat{\mathbf{z}}_a - \hat{\mathbf{z}}_a^{\text{rel}} \right\|^2 - w_{\text{vel}}^{\text{rel}} \sum_{j \in \mathcal{J}_a} \left| \dot{q}_j \right| + w_l L + w_u^{\text{rel}} U$$
- Where: $L = min(\|q_{act.} - q_{lim}^{upper}\| \, , \|q_{act.} - q_{lim}^{lower}\|)$, $U = \|u\|^2$

RESULTS

- The integrated MuJoCo MPC solver took ***too long*** to compute ***for real-time*** control on an: i7 11th Gen CPU with a 3070 Ti GPU machine.
- ***MPC*** control proved ***effective*** for ball catching and end-effector return to initial position.
- ***MPC*** control ***succeeded*** in swirling the ball in the bowl without unintended launching, yet ***frequently failed*** to launch in the intended trajectory.
- ***Energy-shaping*** control resulted in more human-like, fluid motions. It ***consistently*** launched the ball through the intended arcs.
- ***MPC*** also succeeded in generating ***target trajectories*** for the ball to be thrown in scenarios where ***obstacles*** were placed between the arms.
- In the future we may try to use ***reinforcement learning*** for cost shaping to ***refine phase cost weights*** and reduce hand-tuned weights, and ***add disturbance estimation*** and feedback (e.g., ball-tracking using vision) so the juggling becomes ***more robust*** against noisy dynamics and external perturbations.
- Example juggling at ***QR code*** on bottom-left.
- Example ***throwing costs*** over time for both MPC and Energy-shaping when juggling one ball are shown below: